AI-Powered Smart Agriculture Monitoring System

Internship project — Trinetra Drones & Robotics Pvt. Ltd. | Submitted by: Anshika Kodukula | June 2026

ORGANIZATION INTERNSHIP

About Trinetra Drones & Robotics

- Provider of drone solutions for agriculture and industry
- Focus on AI-driven precision farming and analytics
- Products: inspection, spraying, mapping and decision-support

Internship Objectives

- Study drone platforms, sensors and flight systems
- Explore computer vision & ML for crop health
- Design and prototype smart farming concepts

Internship Journey — Weeks 1–3

Week 1 — Foundations

Drone basics, flight controllers, sensors (RGB, multispectral, thermal) and precision farming principles

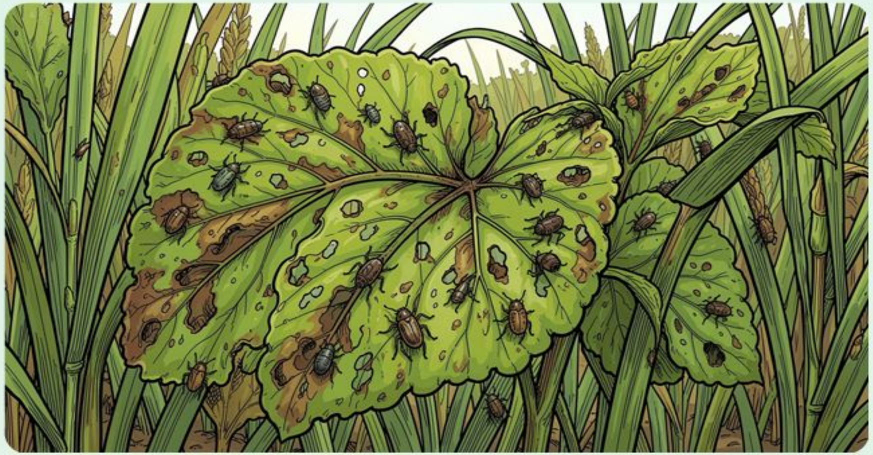
Week 2 — Data & Processing

Collecting agricultural datasets, image preprocessing, annotation and ML fundamentals

Week 3 — Models & Cases

Training disease detection models, transfer learning with CNNs, and studying industry case studies

Agricultural Challenges (Context)



Crop Diseases

Early detection is limited by manual scouting frequency and coverage

Pest Infestations

Pests spread rapidly; delayed response increases losses

Water Scarcity

Inefficient irrigation causes yield reduction and resource waste

Labour & Scale

Shortages and large farm areas limit continuous monitoring

Proposed Solution — System Overview

End-to-end system combining aerial data capture, cloud storage, AI-based analytics and a farmer-facing dashboard for real-time decision support.

Agricultural Drones

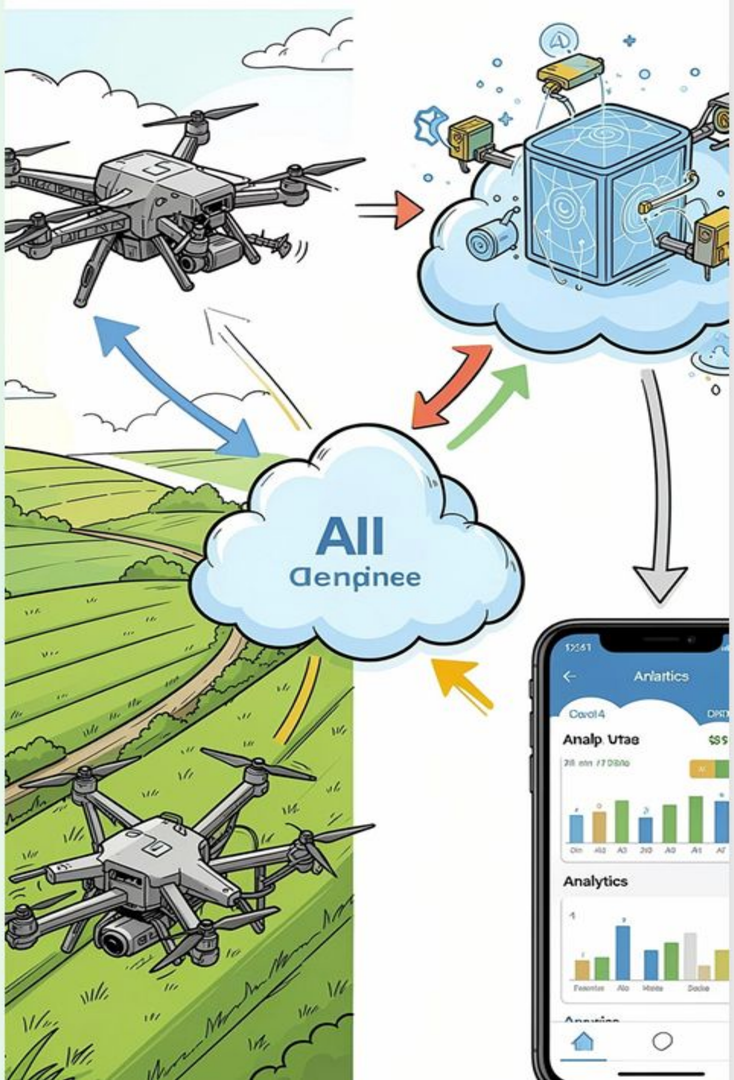
Autonomous flight missions, multispectral and RGB imaging, precision spraying capability

AI Processing Engine

Image segmentation, disease classification, pest detection and yield estimation

Cloud & Storage

Secure data lake, scalable model training and batch/stream processing



System Architecture

- Drone Capture
- Edge Preprocessing
- Cloud AI
- Farmer Dashboard

Key Features & Benefits

Crop Health Monitoring

Field-wide vegetative indices (NDVI, SAVI) highlight stress zones for targeted intervention

Disease & Pest Detection

Deep learning models classify common diseases and pests with confidence scores

Smart Irrigation

Soil moisture mapping and zone-based irrigation scheduling to conserve water

Yield Prediction

Combines imagery, weather and historical yields to forecast production and plan resources

Farmer Dashboard

Actionable alerts, field maps, treatment suggestions and simple mobile workflows

Benefits: reduced crop losses, improved yields, cost savings and sustainable resource use.

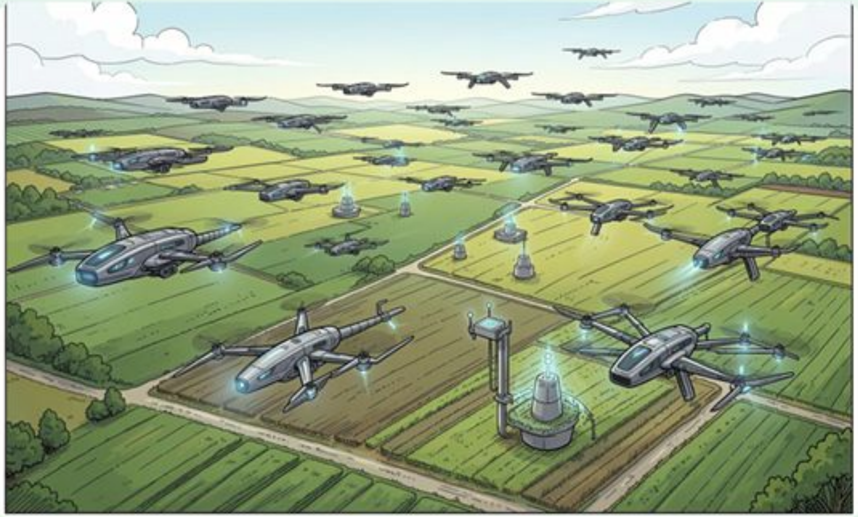
Operational Use Cases & Future Trends

Future Trends

Autonomous drone operations, swarm coordination, edge AI for low-latency inference, IoT sensor fusion, and agricultural robotics.

Use Cases — Drone Solutions

- Precision spraying for targeted pest control
- Automated surveillance for early disease hotspots
- Irrigation monitoring and zone-based prescriptions
- Rapid damage assessment after extreme weather



Social Impact & Outreach

Medical Delivery

Rapid transport of supplies to remote villages; platform reuse for emergencies

Disaster & Fire Detection

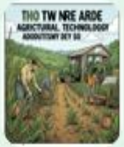
Early identification of fires and flood damage using thermal and optical sensors

Wildlife & Forest Monitoring

Non-intrusive monitoring for conservation and anti-poaching efforts

Digital Marketing Contributions

Promotional posters and corporate video assets to drive farmer adoption



Social media campaigns and educational content to build brand trust

Outcomes, Recommendations & Conclusion

Key Outcomes

- Design and prototype of a Smart Agriculture monitoring system
- Trained disease detection models and curated labelled datasets
- Created marketing materials and social-impact concepts

Recommendations

- Pilot autonomous operations on representative farms
- Deploy advanced edge models for low-latency alerts
- Integrate IoT sensors for sensor fusion and improved accuracy
- Establish farmer training and feedback loops

Conclusion

Combining drones, AI and cloud platforms enables scalable, timely and sustainable improvements in productivity and resource efficiency. Next step: field pilot and farmer co-design.